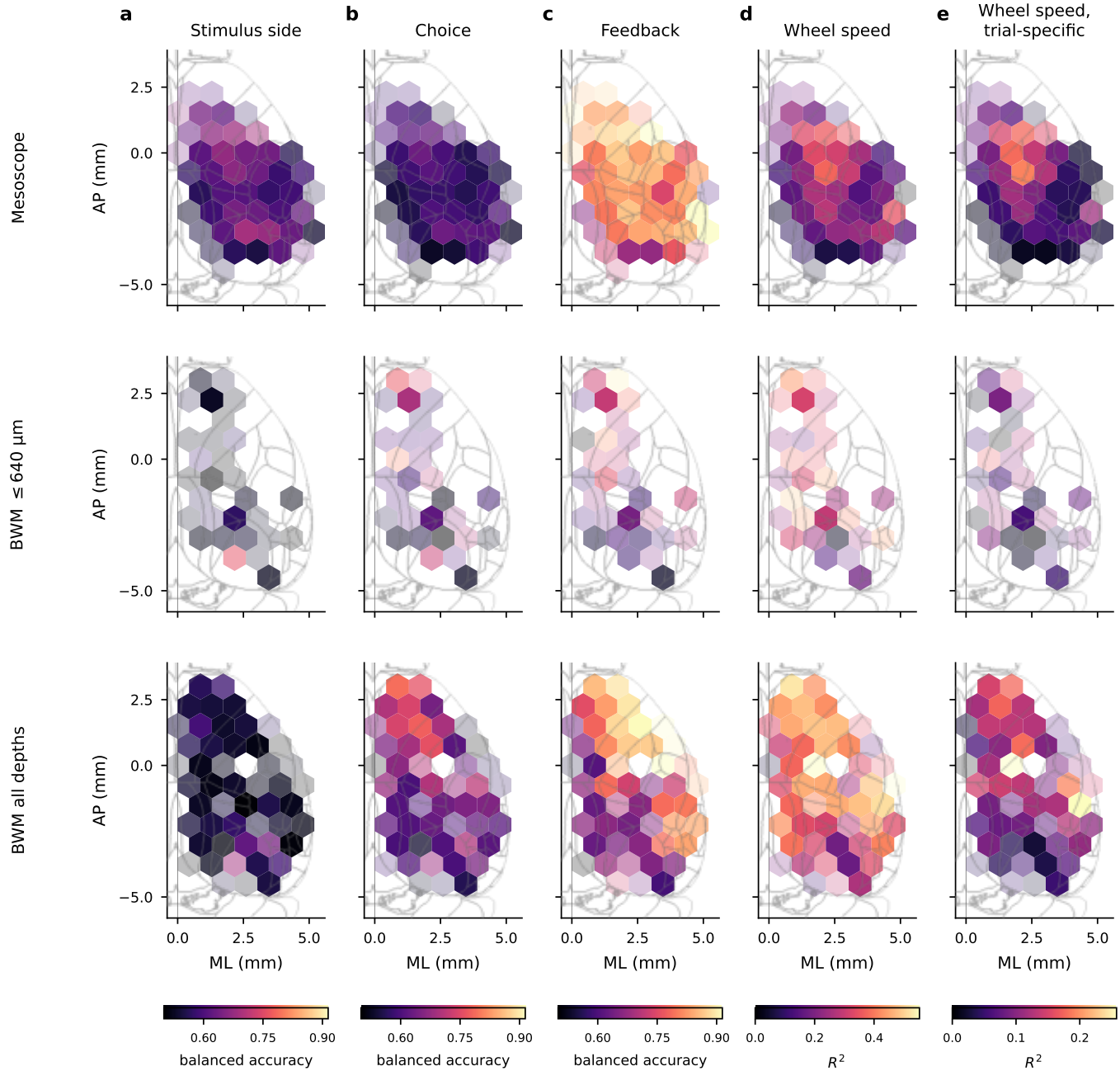


Figure 1: Decoding from all neurons. Rows: mesoscope; BWM units within 640 μm of the pial surface; BWM at all depths. Each bin (1 mm hexagon, hemispheres folded) shows the mean over sessions of decoders trained on all neurons of one session in that bin (sessions with $\geq$20 neurons there); mesoscope bins hold far more neurons than BWM bins. Opacity: number of animals (full at $\geq$4). **a** Stimulus side, balanced accuracy (chance 0.5; colour scale shared by **a**–**c**). **b** Choice. **c** Feedback. **d** Wheel speed, cross-validated R^2. **e** Trial-specific wheel speed, R^2.